

Homework - 4

Building a Mini GPT from Scratch

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Objective. This experiment implemented a decoder-only GPT-style Transformer from scratch in PyTorch and evaluated character-level text generation with greedy, temperature, and top-k decoding. Built-in Transformer and MultiheadAttention modules were not used.

1. Data Preparation

The Shakespeare corpus contained **1,738 characters**. Character-level tokenization produced a vocabulary of **49 unique characters**, represented by `char_to_idx` and `idx_to_char` mappings. A 90/10 split produced 1,564 training tokens and 174 validation tokens.

A sequence length of **128** was used. Each input contains tokens $[t, \dots, t+127]$, while the target is shifted one character forward to $[t+1, \dots, t+128]$. This creates the autoregressive next-character prediction task required for GPT training.

2. Model Architecture

Component	Configuration
Vocabulary size	49
Hidden dimension	128
Attention heads	4
Dimension per head	32
Decoder layers	2
Feedforward dimension	512
Maximum sequence length	128
Total / trainable parameters	425,777 / 425,777

Multi-head masked self-attention was implemented manually with separate query, key, and value linear projections. The hidden representation was split across four heads and a lower-triangular causal mask prevented each position from attending to future characters. Each decoder block contained masked attention, a residual connection and LayerNorm, followed by a Linear-GELU-Linear feedforward network with a second residual connection and LayerNorm.

The complete Mini GPT used token embeddings, learnable positional embeddings, two decoder blocks, final

LayerNorm, and a linear projection from hidden states to the 49-character vocabulary.

3. Training Results

Training used the **Adam** optimizer with learning rate **3e-4**, batch size **64**, cross-entropy loss, and **5 epochs**.

Epoch	Average training loss
1	3.3937
2	2.8129

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Epoch	Average training loss
3	2.5608
4	2.3909
5	2.2806

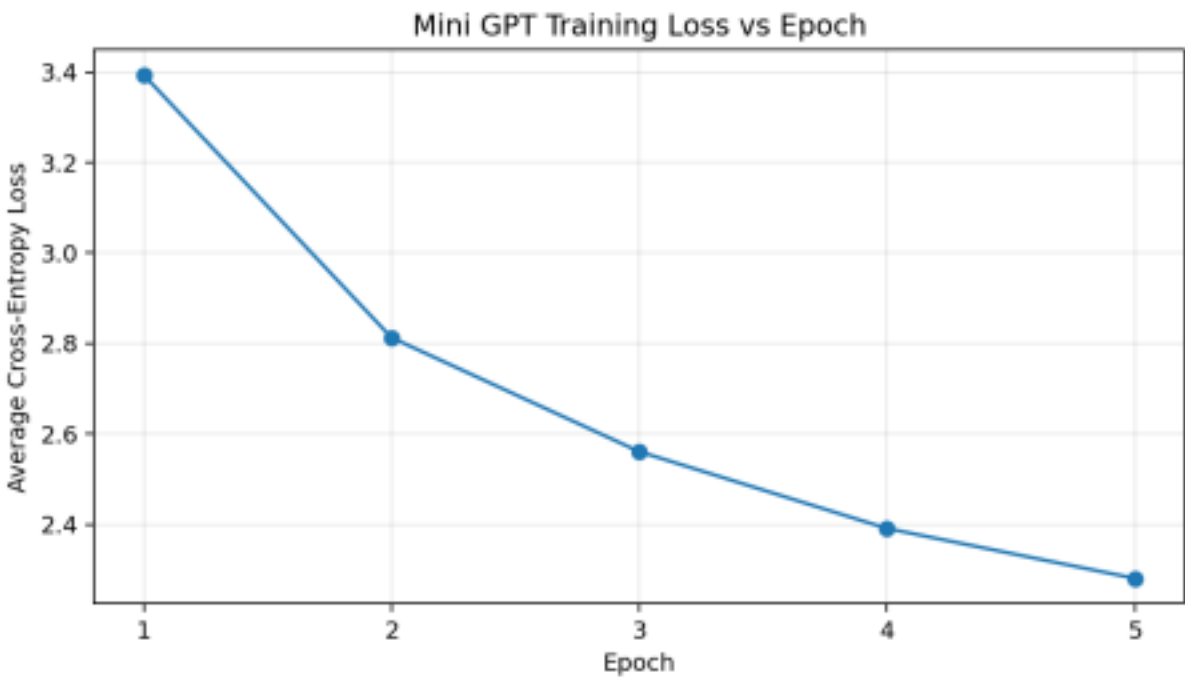


Figure 1. Training loss decreased consistently across the five epochs.

The average training loss fell from **3.3937** in epoch 1 to **2.2806** in epoch 5, an absolute reduction of **1.1131**. The downward trend indicates that the model learned character-level statistical patterns from the training corpus.

4. Sampling Results

All decoding experiments used the prompt **"ROMEO:"** and generated 300 new characters.

Greedy Decoding

Greedy decoding always selected the highest-logit character. It was deterministic but quickly entered a repetitive pattern dominated by fragments such as "the".

ROMEO:\nRor the t the the the the the ar ar the the the the at the the the the athe the
the... **Temperature Sampling**

T = 0.5 produced the most coherent output among the tested settings. It strongly favored high-probability characters and retained recognizable word-like patterns, although repetition remained. **T = 1.0** increased variation, while **T = 1.5** produced the most visibly diverse output but also the least consistent text.

T = 0.5: ROMEO:\nAnd, be t the all t I athatin nathee whe inor f tha thor ar the the
her... T = 1.0: ROMEO:\nI smillfar ar hrswevenathwiny t R?\nHezerther is f wAzengornNa!...
T = 1.5: ROMEO:\nOUJUlloL\nCe. o?'s ?\nW kwhpusmy phomo wSU:NoT:,zheand they!...

Top-k Sampling

k = 5 constrained sampling to a small set of likely characters, producing more repetitive and predictable patterns. **k = 20** expanded the candidate set and increased diversity, but it also introduced more irregular character combinations.

k = 5: ROMEO:\nT:\nHe,e n thanome n s the aitaller stheat amen s athen t sh the
ame... k = 20: ROMEO:\no te, ant otit\nWhhe he ondour met ks besio! Iher thy flt are
khe...

5. Comparison and Findings

Method	Observed behavior	Coherence / diversity
Greedy	Highest-probability character every step; became repetitive.	Deterministic; least diverse.
Temperature 0.5	Sharper distribution; favors likely characters.	Most coherent among tested out re
Temperature 1.0	Moderate randomness and more varied structure.	Greater diversity, lower consistency.
Temperature 1.5	Much flatter distribution and unusual combinations.	Most diverse; lowest coherence.
Top-k 5	Samples only among five strongest candidates.	Constrained and relatively predictable.
Top-k 20	Samples among a larger candidate set.	More diverse than k=5; less consistent.

Coherence: Temperature sampling at 0.5 produced the most coherent output overall in these experiments because it favored high-probability characters while still allowing limited variation. Greedy decoding was deterministic, but its repeated high-probability loop reduced practical coherence.

Diversity: Temperature sampling at 1.5 produced the most diverse output. Increasing temperature flattened the probability distribution and made lower-probability characters more likely to be selected, but this diversity came at the cost of coherence.

Overall finding: The experiment demonstrates the expected coherence-diversity trade-off. Lower-randomness decoding produces more constrained character patterns, while higher-randomness decoding increases variation. The model learned useful character-level structure, as shown by the steady loss decrease, but its generated text remained only partially meaningful because the corpus was very small and training was limited to five epochs.

6. Conclusion

The assignment successfully demonstrates the core mechanics of a GPT-style decoder: character tokenization, causal multi-head self-attention, stacked decoder blocks, autoregressive training, and multiple decoding strategies. The manually implemented four-head attention respected causal masking, and the 425,777-parameter model learned sufficiently to reduce training loss across all five epochs. Sampling experiments showed that decoding strategy has a major effect on generated text: conservative sampling improves consistency, while higher temperature or a larger candidate set increases diversity.

